

Understanding the Problem

I approached understanding the challenge by identifying that the goal was to create a word guessing game that reads words from a file and randomly selects one for the user to guess. The program also needed to track guesses and update the displayed word as the user guesses letters.

At first, I found it confusing how to properly read from a file and store multiple words. I also wasn't sure how to randomly select a word from the file. I resolved this by using an `ArrayList` to store all the words and then using a `Random` object to select one.

Code reference:

```
60 ArrayList<String> wordList = new ArrayList<>();
61 String line;
62
63 // Get the number of words in the file
64 while ((line = readFile.readLine()) != null)
65 {
66     wordList.add(line.toLowerCase());
67 }
68
69 numWords = wordList.size();
70
71 // update the word to guess randomly
72 wordToGuess = rand.nextInt(numWords);
73
74 // iterate to get the secret word
75 secretWord = wordList.get(wordToGuess);
```

Planning the Solution

Before coding, I broke the problem into smaller steps: reading the file, storing words, choosing a random word, and building the guessing game.

I chose `ArrayList` because it makes it easy to store and access words. I used `BufferedReader` for reading the file and `Random` to select a word.

Code reference:

```
58 readFile = new BufferedReader(in);
Random rand = new Random();
```

Implementation

I wrote the code in smaller sections. First, I handled file reading. Then I added the random word selection. Finally, I built the game loop that lets the user guess letters.

I tested the program by running it with different files and checking if the word updates correctly after each guess.

Code reference:

```
04- do {
05     // prompt for a letter
06     System.out.print("Guess a letter (or ! to quit): ");
07     letterGuess = input.nextLine().toLowerCase();

114     /* increment number of guesses */
115     numGuesses++;
116
117     /* update word */
118     StringBuilder temp = new StringBuilder(wordSoFar);
119
120-    for (int i = 0; i < secretWord.length(); i++)
121    {
122-        if (secretWord.charAt(i) == letterGuess.charAt(0))
123        {
124            temp.setCharAt(i, letterGuess.charAt(0));
125        }
126    }

128    wordSoFar = temp.toString();
129
130    /* display guessed letter instead of dash */
131    System.out.println("Secret word: " + wordSoFar);
```

Overcoming Challenges

The most difficult part was updating the word correctly when a user guessed a letter. I also found file handling a bit challenging at first.

When I got stuck, I focused on one part at a time and tested it before moving on.

Code reference:

```
48- try {
49     // prompt user for file name
50     System.out.print("Enter the name of the word file: ");
51     fileName = input.nextLine();
52
53     // initialize the file reader object to name of the file object
54     wordFile = new File(fileName);
55     in = new FileReader(wordFile);
```

Learning

I learned how to read from files, use ArrayList, and build an interactive program. I also improved my understanding of loops and string manipulation.

This will help me in future programming tasks involving user input and file processing.

Code reference:

```
101      System.out.println("Secret word: " + wordSoFar);

136     if (wordSoFar.equals(secretWord))
137     {
138         System.out.println("Congratulations! You guessed it!");
139     }
```